Unit 9: Convolutional Neural Networks (CNNs)

1. What?

This unit introduced the **basic concepts of Convolutional Neural Networks (CNNs)**, a specialized form of Artificial Neural Networks (ANNs) designed for image and visual data processing. Unlike traditional ANNs, CNNs can automatically detect important patterns and features within images.

Key concepts covered included:

- **Three main stages of CNN processing:**
  - **Filtering (Convolution):** Applying kernels/filters to extract useful image features.
  - **Detection:** Using activation functions such as **ReLU (Rectified Linear Unit)** to determine whether detected features should be activated.
  - **Condensation:** Reducing dimensionality through pooling while preserving important information.
- **Kernels, Filters, and Feature Maps**
  - A **kernel (or filter)** is a small matrix that scans across an image.
  - The filter extracts specific patterns such as edges, lines, or textures.
  - The output produced after applying a filter is called a **feature map**.
- **Activation Functions**
  - CNNs use activation functions, particularly **ReLU**, to introduce non-linearity and allow the network to learn complex patterns.
- **Pooling and Flattening**
  - **Pooling layers** reduce the size of feature maps and computational requirements while retaining key information.
  - **Flattening** converts the multidimensional output into a one-dimensional vector that can be processed by fully connected layers for classification.

Through these concepts, I gained a deeper understanding of how CNNs transform raw image data into meaningful features that can be used for accurate predictions and classification tasks.

---

2. So What?

This unit was particularly exciting because it moved beyond theory and showed how CNNs are implemented in practice using **Python and TensorFlow**.

Some important practical insights extracted from the e-portfolio activity are:

- **Image Scaling and Normalization**

    - Scaling pixel values helps improve training efficiency and model performance by creating more consistent input data.
- **One-Hot Encoding**
    - One-hot encoding is essential for multi-class classification problems because it converts categorical labels into a format that neural networks can process effectively.
- **Multiple Convolutional Layers**
    - I learned why a second convolutional layer is often necessary, especially when working with larger images and multiple color channels. Additional layers allow the network to learn increasingly complex features, from simple edges to more sophisticated shapes and objects.
- **Choosing Filters and Kernel Sizes**
    - The number of filters and kernel dimensions significantly impact model performance.
    - Common practices such as using **3×3 kernels** and gradually increasing the number of filters help balance feature extraction and computational efficiency.

Seeing how these concepts worked together in TensorFlow helped bridge the gap between theory and real-world machine learning applications. Building and training CNN models gave me a clearer understanding of how modern image recognition systems operate.

3. Now What? — Learning, Change, and Future Application

This unit has significantly expanded my understanding of how deep learning models process visual information. Before learning about CNNs, I understood image recognition primarily as a machine learning task; however, I now appreciate how CNNs automatically learn hierarchical features, progressing from simple patterns such as edges and textures to more complex objects and shapes. This has changed the way I think about computer vision and demonstrated why CNNs have become the foundation of many modern image-analysis systems.

The results of the experiment attached to my e-portfolio show that adding a dropout layer produced only a modest increase in overall performance. However, the improvement was consistently reflected across precision, recall, and F1-score, indicating that the model became more balanced in its predictions rather than simply improving a single metric. This demonstrates the effectiveness of dropout as a regularization technique, as it reduced overfitting and improved the model's ability to generalize to unseen data. Moving forward, I would explore additional regularization methods, such as data augmentation, batch normalization, or further tuning of dropout rates and network architecture, to achieve greater improvements in test performance while maintaining strong generalization.

As a result of this learning, I have gained greater confidence in working with neural networks and implementing deep learning solutions using TensorFlow. The hands-on experience of preparing image data, applying one-hot encoding, configuring convolution and pooling layers, and training CNN models has helped me move beyond theoretical knowledge to practical application. I now recognize the importance of model architecture decisions, data preprocessing, and parameter selection in achieving effective performance.

Looking ahead, I plan to continue developing my skills in computer vision by exploring more advanced CNN architectures and techniques. I would like to experiment with methods such as data augmentation, transfer learning, dropout, and batch normalization to improve model accuracy and generalization. Additionally, I aim to build more independent projects that involve image classification and object recognition so that I can strengthen both my technical skills and problem-solving abilities.

In my future academic and professional work, I expect CNNs to be valuable tools for solving real-world problems involving visual data. Applications such as medical image analysis, quality inspection in manufacturing, autonomous systems, and facial or object recognition demonstrate the broad impact of CNN technology. By continuing to build my knowledge and practical experience, I hope to leverage deep learning techniques to develop innovative solutions and contribute to projects that require intelligent image analysis and decision-making.